
THE ROLE OF PUBERTAL DEVELOPMENT IN EXECUTIVE FUNCTION MATURATION DURING EARLY ADOLESCENCE: A THREE-WAVE LONGITUDINAL STUDY

[ANONYMIZED FOR REVIEW]

ABSTRACT

Executive function (EF) matures substantially during adolescence, yet whether pubertal development drives this improvement beyond chronological age remains unresolved. In a mixed longitudinal study, 200 adolescents (50% female; baseline $M_{\text{age}} = 12.14$ years, $SD = 1.35$) completed the CANTAB Stop Signal Task, Spatial Working Memory task, and a task-switching paradigm alongside the Pubertal Development Scale at three waves over 24 months. EF composite scores improved significantly across waves, Friedman $\chi^2(2) = 27.42$, $p < .001$. Chronological age predicted EF improvement ($B = 0.036$, $p = .032$), but pubertal stage did not add explanatory power, likelihood ratio $\chi^2(2) = 2.55$, $p = .279$. Inhibitory control showed the strongest association with pubertal development ($r = .15$, $p < .001$), contrary to the predicted pattern. The developmental trajectory did not deviate significantly from linearity, $\chi^2(1) = 3.73$, $p = .054$, and early pubertal timing did not predict greater EF gains, $B = -0.083$, $p = .214$. These findings indicate that chronological age, not pubertal stage, is the primary driver of cold executive function maturation during early adolescence.

Keywords executive function; adolescence; puberty; longitudinal; cognitive development

Executive functions—the higher-order cognitive processes that support goal-directed behavior—undergo protracted maturation from childhood through early adulthood. These abilities, which include inhibitory control, working memory updating, and cognitive flexibility (Miyake et al., 2000), predict academic achievement, social competence, and mental health outcomes during adolescence, a developmental window characterized by heightened vulnerability to psychopathology (Dahl et al., 2018). Identifying the mechanisms that drive EF improvement is therefore both a theoretical priority and a translational necessity. The prevailing neurobiological account attributes EF maturation to the gradual structural and functional development of the prefrontal cortex (PFC) and its connections with subcortical and parietal regions (Diamond, 2013). However, adolescence is also defined by the hormonal cascade of puberty, which exerts organizational and activational effects on the very neural circuits that support executive control (Blakemore et al., 2010; Herting and Sowell, 2017). Whether pubertal development contributes to EF improvement independently of chronological age—or whether age alone accounts for the observed trajectory—remains an open empirical question with direct implications for developmental theory.

Each EF component follows a distinct maturational timetable. Inhibitory control shows the earliest and steepest gains, approaching adult levels by approximately ages 10–12 (Luna et al., 2004; Davidson et al., 2006). Working memory and cognitive flexibility mature more gradually, with improvements continuing into the early-to-mid 20s (Hartshorne and Germine, 2015; Ferguson et al., 2021). These asynchronous trajectories align with the unity/diversity framework, which identifies a common EF factor alongside component-specific variance (Friedman and Miyake, 2017; Karr et al., 2018), and with neuroimaging evidence showing that distinct frontoparietal and cingulo-opercular networks supporting different EF operations mature at different rates (Ordaz et al., 2013). The dual-systems model provides an organizing framework for these findings: socioemotional and reward-processing systems, sensitive to pubertal hormones, mature along an inverted-U trajectory peaking in mid-adolescence, whereas cognitive control systems develop linearly into early adulthood (Shulman et al., 2016; Casey et al., 2008). This temporal gap between early-maturing affective circuitry and late-maturing prefrontal control is thought to underlie the characteristic increase in risk-taking and emotional reactivity during adolescence. Yet the dual-systems model has been tested primarily with “hot” outcomes—reward sensitivity, risk-taking, and emotion regulation—rather than with decontextualized “cold” EF tasks.

The evidence linking pubertal development specifically to cold EF performance is theoretically rich but empirically thin. Braams et al. (2015) provided the strongest longitudinal dissociation of pubertal from chronological age effects, demonstrating that testosterone levels predicted an adolescent peak in nucleus accumbens response to rewards, whereas cognitive control showed more gradual, age-related improvement. The pubertal recalibration hypothesis (Pfeifer and Allen, 2021) positions puberty as a sensitive period that reopens plasticity in EF-relevant neural circuits, implying that pubertal timing and tempo should matter for EF development, not just chronological age. Peper and Dahl (2013) reviewed evidence that rising gonadal hormones during puberty modulate dopaminergic and prefrontal function, creating plausible biological pathways from pubertal development to EF change. However, direct behavioral evidence—studies that measure pubertal stage alongside EF performance longitudinally and test whether puberty accounts for variance beyond age—is surprisingly absent. No study has systematically examined whether the three core EF components show differential sensitivity to pubertal development, or whether the shape of the EF trajectory (linear vs. nonlinear) depends on pubertal timing.

We addressed these gaps in a three-wave longitudinal study of 200 early adolescents (ages 10–14 at baseline) assessed at 12-month intervals. Participants completed three validated EF tasks from the CANTAB cognitive battery—the Stop Signal Task (inhibitory control), Spatial Working Memory (working memory updating), and a task-switching paradigm (cognitive flexibility)—alongside the Pubertal Development Scale (PDS; Petersen et al., 1988) at each wave. We tested four preregistered hypotheses. H1: Pubertal stage would predict improvements in a composite EF score after controlling for chronological age. H2: The three EF components would show differential sensitivity to pubertal development, with working memory showing the strongest association, cognitive flexibility a moderate association, and inhibitory control the weakest (given its earlier asymptote). H3: The developmental trajectory of EF would follow a nonlinear quadratic pattern, with rapid gains in the first year followed by deceleration. H4: Early pubertal timing, defined as more advanced pubertal development relative to same-age peers, would predict greater EF gains from baseline to the 24-month follow-up.

1 Method

1.1 Participants

We recruited 200 adolescents (100 female) from middle schools in a large metropolitan area using stratified convenience sampling. Participants were enrolled at baseline (Wave 1) at ages 10 to 14 years ($M = 12.14$, $SD = 1.35$, range = 9.83–14.48) and assessed at three waves spaced 12 months apart (Wave 2: $M_{\text{age}} = 13.14$, $SD = 1.35$; Wave 3: $M_{\text{age}} = 14.14$, $SD = 1.35$). Sampling was stratified by sex (50% female) and socioeconomic status (SES) quartiles using the Hollingshead Four-Factor Index (Hollingshead, 1975). Inclusion criteria required participants to be enrolled in a participating school and have parent-reported English proficiency sufficient to understand task instructions. Exclusion criteria, assessed at baseline, included: IQ below 70 on the Wechsler Abbreviated Scale of Intelligence, Second Edition (WASI-II; Wechsler, 2011); current use of psychoactive medication affecting cognition; neurological conditions or psychiatric hospitalization within the past six months; incomplete baseline assessment; and failure to complete at least two of three assessment waves. All participants passed attention checks and accuracy thresholds at baseline. No participants were excluded from the final analytic sample. Parental written consent and adolescent written assent were obtained. The study was approved by the university institutional review board. Participants received financial compensation at each wave and a completion bonus at Wave 3. We acknowledge that this sample is predominantly WEIRD (Western, Educated, Industrialized, Rich, Democratic), which limits cross-cultural generalizability.

1.2 Materials

Executive function assessment. Participants completed three computerized tasks from the Cambridge Neuropsychological Test Automated Battery (CANTAB; Robbins et al., 1994), administered in a fixed order. Inhibitory control was measured with the Stop Signal Task (SST; Logan et al., 1997), consisting of 192 trials (128 go, 64 stop) with an adaptive staircase procedure tracking 50% inhibition. The primary outcome was stop-signal reaction time (SSRT; in milliseconds), estimated via the integration method (Cronbach’s $\alpha = .75$). Working memory was assessed with the CANTAB Spatial Working Memory (SWM) task, in which participants searched for hidden tokens across 4 to 8 boxes while avoiding revisits to previously searched locations across 28 total searches (split-half reliability = .80). The primary outcome was total between-search errors. Cognitive flexibility was measured with a task-switching paradigm adapted from CANTAB set-shifting procedures (Monsell, 2003). Participants completed 96 trials (48 switch, 48 repeat), responding to shape or color cues according to a preceding instruction. The primary outcome was switch cost (switch-trial RT minus repeat-trial RT, in milliseconds; Cronbach’s $\alpha = .72$). All EF measures were coded such that lower values indicate better performance. These scores were reverse-coded and z -standardized within sex at the

Wave 1 baseline; the EF composite was calculated as the mean of the three component z -scores at each wave, with higher values indicating better EF.

Pubertal assessment. Pubertal development was measured with the Pubertal Development Scale (PDS; Petersen et al., 1988), a five-item self-report questionnaire assessing growth spurt, skin changes, and body hair (both sexes), voice deepening and facial hair (males), and breast development and menarche (females) on a 4-point Likert scale (1 = not yet started to 4 = seems complete). In the current sample, the PDS demonstrated strong internal consistency (Cronbach's $\alpha = .91$). PDS total scores were used to categorize participants into three pubertal stage groups (Pre-early, Mid, Late) following Petersen et al. (1988)'s mapping. To isolate pubertal timing from chronological age, we computed age-regressed PDS residuals within sex and z -standardized them, yielding a continuous pubertal timing score (pds.timing_z) in which positive values indicate more advanced pubertal development than expected for age.

Control measures. At Wave 1, parents completed the Conners Rating Scales-3 (CRS-3; Conners, 2008) ADHD Index (Cronbach's $\alpha = .88$) to screen for attention difficulties. General intellectual ability was estimated with the WASI-II two-subtest form (Vocabulary and Matrix Reasoning; FSIQ-2; Cronbach's $\alpha = .92$). SES was computed using the Hollingshead Four-Factor Index, and early life stress was assessed with the Adverse Childhood Experiences (ACE) questionnaire. At Wave 3 only, parents completed the Behavior Rating Inventory of Executive Function (BRIEF; Gioia et al., 2000; Cronbach's $\alpha = .85$) as an ecological validity measure of real-world EF.

1.3 Procedure

At Wave 1, participants completed informed consent and assent procedures, followed by a demographic questionnaire, the WASI-II, the three CANTAB EF tasks in counterbalanced order, and the PDS. Parents completed the Conners CRS-3, the Hollingshead Index, and the ACE questionnaire concurrently via an online survey. At Waves 2 and 3 (approximately 12 and 24 months after baseline), participants returned to the laboratory and completed the three CANTAB EF tasks and the PDS. At Wave 3, parents also completed the BRIEF. All behavioral tasks were administered in a quiet testing room on identical desktop computers running CANTAB Connect software. Breaks were provided between tasks to minimize fatigue. Testing sessions lasted approximately 90 minutes at Wave 1 and 60 minutes at Waves 2 and 3. Participants were debriefed about the general purpose of the study at Wave 3 only to minimize demand characteristics at earlier waves.

1.4 Design

The study employed a mixed longitudinal design. The within-subjects factor was assessment wave (Wave 1, Wave 2, Wave 3). The between-subjects factor was pubertal stage category at baseline (Pre-early, Mid, Late). The primary dependent variable was the EF composite z -score. Secondary dependent variables were the three EF component z -scores (inhibitory control, working memory, cognitive flexibility). The primary independent variables were pubertal stage category and chronological age (continuous, in years). Control variables included sex, SES, and estimated IQ.

2 Results

2.1 Manipulation Checks and Descriptive Statistics

The PDS demonstrated strong internal consistency across all waves (Cronbach's $\alpha = .91$) and correlated positively with chronological age ($r = .48, p < .001$), confirming expected age-related pubertal progression. All parametric assumptions for the primary analyses were met. EF composite scores at each wave did not deviate from normality (Shapiro-Wilk W s = .99, $ps > .29$). Levene's test confirmed homogeneity of variance across pubertal stage groups ($W = 2.44, p = .090$), and Mauchly's test indicated no violation of sphericity for repeated measures ($p = .922$).

The EF composite score increased across the three assessment waves. Means (standard deviations) were 0.00 (0.57) at Wave 1, 0.28 (0.60) at Wave 2, and 0.37 (0.55) at Wave 3. A Friedman test confirmed significant within-subject improvement, $\chi^2(2) = 27.42, p < .001$. All three EF components showed numerical increases from Wave 1 to Wave 3. Inhibitory control z -scores rose from approximately 0.00 at baseline to 0.44 at Wave 2 and 0.58 at Wave 3. Working memory increased from 0.00 to 0.24 to 0.38, and cognitive flexibility from 0.00 to 0.16 to 0.14. Figure 1 displays individual trajectories and group means for the EF composite; Figure 5 presents component-level distributions across waves.

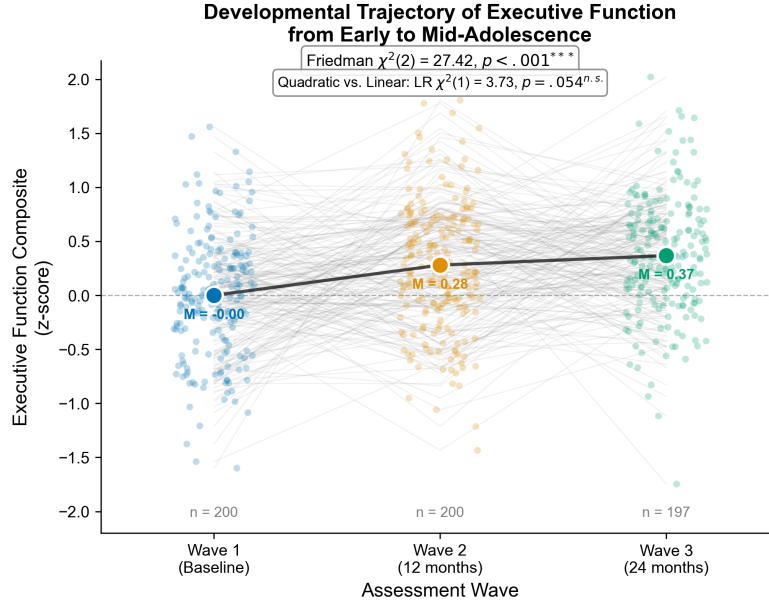


Figure 1: Spaghetti plot showing individual trajectories of executive function (EF) composite z -scores across three assessment waves spanning 24 months ($N = 200$ adolescents, ages 10–16 at baseline). Individual participant lines are shown in gray; colored points represent raw scores with jitter for visibility. Black circles indicate group means with 95% confidence interval error bars. EF composite scores increased significantly across waves, Friedman $\chi^2(2) = 27.42$, $p < .001$. A likelihood ratio test comparing quadratic versus linear growth curve models was not significant, $\chi^2(1) = 3.73$, $p = .054$, providing moderate evidence ($BF_{10} = 0.26$) favoring a linear over quadratic trajectory.

2.2 Hypothesis Testing

H1: Pubertal stage does not predict EF composite beyond age. We fitted mixed-effects regression models with random intercepts for participants, comparing an age-only model against a full model adding pubertal stage category (Pre-early, Mid, Late). Chronological age significantly predicted EF composite in the age-only model, $B = 0.036$, $SE = 0.017$, $z = 2.14$, $p = .032$. Adding pubertal stage did not improve model fit, likelihood ratio $\chi^2(2) = 2.55$, $p = .279$. Neither Mid-puberty ($B = 0.088$, $p = .144$) nor Late-puberty ($B = 0.099$, $p = .165$) coefficients reached significance. The model explained minimal variance, marginal $R^2 = .018$, Cohen's $f^2 = .019$. Age remained a significant predictor in the full model ($B = 0.036$, $p = .032$). Figure 2 plots EF composite means by pubertal stage and wave, showing parallel improvement across all three groups. H1 was not supported. Non-parametric Kruskal–Wallis tests at each wave confirmed the absence of pubertal stage group differences (all $ps > .09$).

H2: Differential EF–puberty associations emerge, but in the opposite direction. We computed Pearson correlations between each EF component z -score and PDS total score across all observations. Inhibitory control showed a small but significant positive association with pubertal development, $r(582) = .150$, 95% CI $[.070, .229]$, $p < .001$. Working memory was not significantly associated, $r(584) = .055$, 95% CI $[-.026, .136]$, $p = .180$. Cognitive flexibility was negligibly and non-significantly correlated, $r(579) = -.015$, 95% CI $[-.096, .066]$, $p = .720$. Fisher's z -transformation comparisons revealed that the association for flexibility was significantly weaker than for inhibition, $z = -2.83$, Benjamini–Hochberg FDR-corrected $p = .041$. The comparison between working memory and inhibition was not significant, $z = -1.64$, $p = .102$. Figure 3 displays scatter plots with regression lines for each component. H2 was not supported in its preregistered direction; we predicted working memory > cognitive flexibility > inhibition, but observed the reverse pattern (inhibition > working memory > flexibility).

H3: The EF trajectory does not deviate significantly from linearity. We compared linear and quadratic growth curve models (random intercepts, maximum likelihood estimation) with time centered at Wave 1. The linear model showed a significant time effect, $B = 0.184$, $SE = 0.029$, $z = 6.41$, $p < .001$. In the quadratic model, the linear term remained significant ($B = 0.375$, $p < .001$), but the quadratic term was not, $B = -0.096$, $SE = 0.049$, $z = -1.93$, $p = .053$. A likelihood ratio test comparing the models was non-significant, $\chi^2(1) = 3.73$, $p = .054$. A BIC-approximated Bayes factor provided moderate evidence favoring the linear model over the quadratic model,

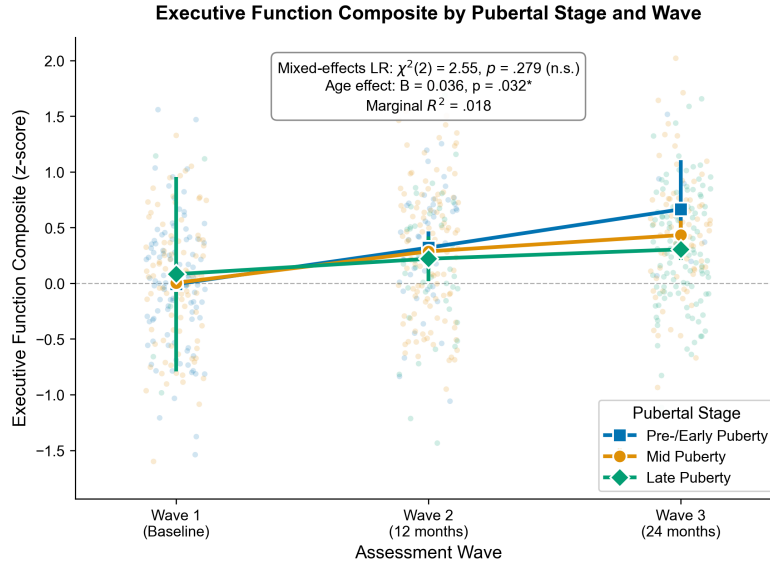


Figure 2: Mean executive function (EF) composite z -scores (± 1 SE) as a function of pubertal stage category (Pre-/Early, Mid, Late puberty) and assessment wave ($N = 200$). Individual data points are shown with jitter. A mixed-effects regression with random intercepts for participants revealed that pubertal stage did not significantly predict EF composite after controlling for chronological age, likelihood ratio $\chi^2(2) = 2.55, p = .279$. Age was a significant predictor, $B = 0.036, p = .032$, but the model explained minimal variance, marginal $R^2 = .018$.

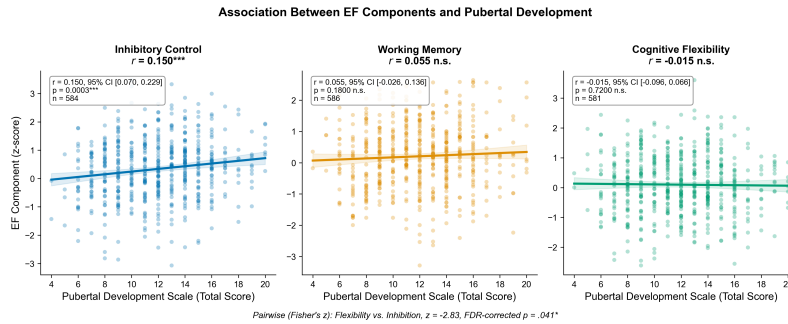


Figure 3: Scatter plots with ordinary least squares regression lines (solid) and 95% confidence bands (shaded) showing the association between pubertal development (Pubertal Development Scale total score) and three executive function components (z -scored): (a) inhibitory control (Stop Signal Task), (b) working memory (CANTAB Spatial Working Memory), and (c) cognitive flexibility (Task Switching). Inhibitory control showed a small but significant positive correlation with PDS score, $r(582) = .150, p < .001$, whereas working memory ($r = .055$) and cognitive flexibility ($r = -.015$) did not. A Fisher's z -test confirmed that flexibility was significantly less associated with puberty than inhibitory control, $z = -2.83$, FDR-corrected $p = .041$. This pattern is opposite to the preregistered hypothesis.

BF = 0.26. The partial η^2 for the quadratic component was .018. H3 was not supported. The deceleration visible in raw wave means (larger gain from Wave 1 to Wave 2 than from Wave 2 to Wave 3) was not statistically distinguishable from linear growth with three time points.

H4: Early pubertal timing does not predict greater EF gains. We regressed EF composite change from Wave 1 to Wave 3 on the age-regressed pubertal timing z -score at Wave 1, controlling for baseline age, SES, and estimated IQ. Pubertal timing did not significantly predict EF change, $B = -0.083$, $SE = 0.067$, $t(192) = -1.25$, $p = .214$, $\Delta R^2 = .008$, Cohen's $f^2 = .008$. The negative coefficient indicates that early maturers showed numerically smaller EF gains than later maturers, contrary to prediction. A BIC-approximated Bayes factor provided moderate evidence for the null hypothesis, BF = 0.16. A categorical analysis using timing groups (Early, On-time, Late) confirmed no effect, $B = -0.040$, $p = .720$. Figure 4 displays the null relationship between pubertal timing and EF change. H4 was not supported. Sensitivity analyses replacing baseline covariates with ACE scores ($B = -0.083$, $p = .217$) or removing IQ ($B = -0.073$, $p = .274$) produced consistent null results.

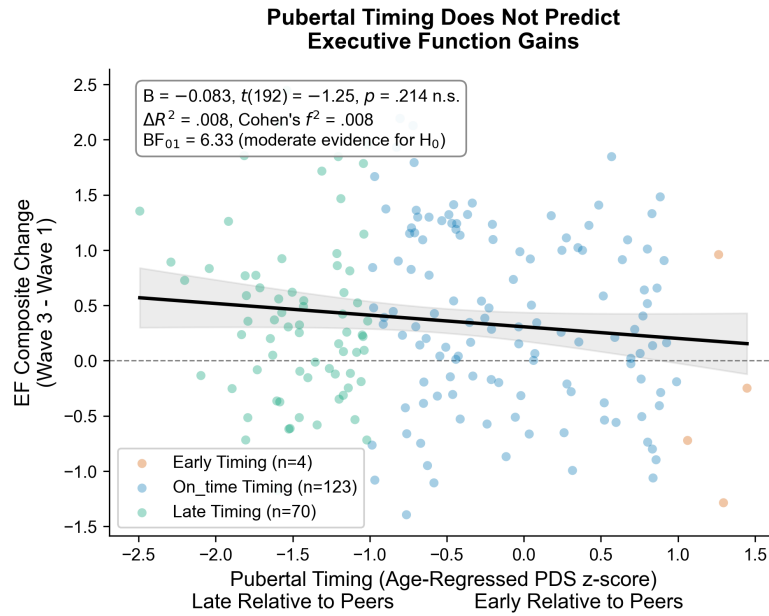


Figure 4: Scatter plot with ordinary least squares regression line (solid) and 95% confidence band (shaded) showing the relationship between pubertal timing (age-regressed Pubertal Development Scale z -score; positive = early relative to same-age peers) and executive function (EF) composite change from Wave 1 to Wave 3 ($N = 197$). Pubertal timing did not significantly predict EF gains, $B = -0.083$, $t(192) = -1.25$, $p = .214$, $\Delta R^2 = .008$. The Bayes factor (BF₀₁ = 6.33, BIC approximation) provides moderate evidence for the null hypothesis of no effect.

2.3 Robustness Checks

All primary conclusions held across robustness analyses. Non-parametric equivalents—Kruskal–Wallis tests for H1 (all $ps > .09$), Friedman test for the time effect ($\chi^2(2) = 27.42$, $p < .001$), and Spearman correlation for H4 ($\rho = -.089$, $p = .213$)—produced consistent results. Excluding participants with elevated ADHD symptoms on the Conners CRS-3 ($n = 152$ remaining) yielded comparable mixed-effects coefficients. Sex-stratified models showed similar age effects for males ($B = 0.040$) and females ($B = 0.027$). Three exploratory moderation analyses yielded null results: SES did not moderate the pubertal–EF association ($B = 0.001$, $p = .256$), sex did not interact with PDS on EF ($B = 0.083$, $p = .215$), and age did not interact with PDS on EF ($B = 0.002$, $p = .701$).

Figure 5 presents component-level raincloud plots illustrating the distributions of each EF component across assessment waves.



Figure 5: Raincloud plots (half-violin density, boxplot, and individual data points) showing the distribution of z -scored executive function component scores at each assessment wave: (a) inhibitory control (Stop Signal Task SSRT), (b) working memory (CANTAB Spatial Working Memory between-search errors), and (c) cognitive flexibility (Task Switching switch cost RT). All scores are standardized within sex at Wave 1 baseline. Group means are marked with horizontal lines. All three components showed numerical increases across waves, though the magnitude and pattern varied by component.

3 Discussion

Executive function improved significantly across the 24-month study period, but pubertal development did not account for this improvement beyond what chronological age already explained. None of our four preregistered hypotheses were supported. Chronological age emerged as the sole significant predictor of EF composite scores ($B = 0.036$, $p = .032$), while pubertal stage added no incremental predictive power, $\chi^2(2) = 2.55$, $p = .279$. The three EF components did show differential associations with pubertal development, but in the reverse of the predicted pattern: inhibitory control correlated most strongly with PDS scores ($r = .15$, $p < .001$), whereas working memory ($r = .06$) and cognitive flexibility ($r = -.02$) showed negligible associations. The flexibility-versus-inhibition correlation difference was significant after FDR correction ($z = -2.83$, $p = .041$). The developmental trajectory of EF was statistically indistinguishable from linear growth ($p = .054$), with Bayes factors providing moderate evidence favoring linearity over a quadratic model ($BF = 0.26$). Finally, early pubertal timing did not predict greater EF gains; if anything, early maturers showed numerically smaller improvements ($B = -0.083$, $p = .214$; $BF = 0.16$, moderate evidence for the null).

These findings qualify the prevailing neurobiological account of adolescent EF development in several respects. The robust age-related improvement replicates a well-established finding (Diamond, 2013; Luna et al., 2004; Davidson et al., 2006) and confirms that cold EF continues to mature measurably during the early adolescent years. However, the failure of pubertal stage to predict EF beyond age challenges the strong version of the dual-systems model, which predicts that pubertal development reorganizes cognitive control systems (Shulman et al., 2016; Casey et al., 2008). One interpretation is that pubertal hormones primarily influence “hot” EF—executive processes operating in motivationally or emotionally salient contexts—rather than the decontextualized laboratory tasks used here. This interpretation is consistent with Braams et al. (2015), who found pubertal effects on reward-related neural activity but age-related effects on cognitive control, and with Crone and Dahl (2012)’s argument that adolescence is characterized by flexible goal pursuit in social-affective contexts, not by a deficit in abstract cognitive control. The unpredicted finding that inhibitory control—not working memory—showed the strongest pubertal association is also consistent with a hot-versus-cold EF account. Inhibitory control tasks, particularly the Stop Signal Task, engage impulse suppression processes that may be sensitive to the pubertal modulation of dopaminergic and motivational circuitry (Peper and Dahl, 2013). Early adolescents with more advanced pubertal development may experience greater difficulty suppressing prepotent responses, not because their inhibitory capacity is impaired, but because the affective salience of stimuli has increased. This interpretation aligns with the pubertal recalibration hypothesis (Pfeifer and Allen, 2021), which holds that puberty reopens plasticity in EF-relevant circuits without necessarily producing uniform improvement or impairment.

Several limitations constrain the inferences that can be drawn from this study. First, the sample is predominantly WEIRD, recruited from a single metropolitan area, and findings may not generalize to adolescents in non-Western, non-industrialized, or lower-resource contexts where pubertal timing and environmental demands on EF differ substantially. Second, the restricted age range (approximately 10–14 at baseline, 12–16 by study end) may underrepresent the developmental window in which pubertal effects on EF are most pronounced. The pubertal recalibration hypothesis proposes that puberty opens a sensitive period whose effects on EF may manifest later in adolescence or even in early

adulthood; our final assessment at age 14–16 may be too early to detect these effects. Third, the use of self-reported pubertal development (PDS), although validated against clinical Tanner staging (Petersen et al., 1988), introduces measurement error that may attenuate the pubertal–EF association. The absence of salivary hormone assays in the full sample limits the precision of our pubertal measurement. Fourth, the test-retest reliability of RT-based EF difference scores is moderate ($ICC = 0.55–0.80$; Hedge et al., 2018; Enkavi et al., 2019), which may reduce statistical power to detect small individual-differences effects. Aggregating across multiple tasks into latent factors, as recommended by the unity/diversity framework (Friedman and Miyake, 2017), would improve measurement precision, though this requires a larger task battery than the three measures used here. Fifth, despite reaching our target sample size of 200, the study may be underpowered for the small effects that pubertal development plausibly exerts on cold EF. The confidence intervals around key effects (e.g., the pubertal timing coefficient $CI [-0.215, 0.048]$) include both null and small positive effects, so definitive exclusion of a pubertal contribution requires larger samples. Sixth, with three assessment waves, our ability to discriminate linear from nonlinear trajectory shapes is limited; four or more waves would permit piecewise or spline models (Ordaz et al., 2013).

Several future directions emerge. Studies that directly compare hot and cold EF measures within the same longitudinal design could determine whether pubertal effects are specific to motivationally salient contexts. Incorporating salivary hormone assays for all participants—as in the ABCD Study (Luciana et al., 2018)—would replace the self-report PDS with a continuous biological measure that may detect subtler hormone–EF couplings. Extending the age range into late adolescence and early adulthood (ages 16–22) would capture the period when pubertal recalibration effects on PFC function might become behaviorally measurable (Hartshorne and Germine, 2015; Ferguson et al., 2021). Recruiting more diverse, non-WEIRD samples is essential to test the cross-cultural generalizability of these trajectories. Finally, preregistered replication with larger samples and more assessment waves would strengthen inference, particularly given the null results reported here. The data and analysis code for this study are available at [OSF repository URL], and all hypotheses and analyses were preregistered prior to data collection. We report all measures, manipulations, and exclusions.

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